

Lepton Flavour Violation searches in $D_{(s)}^+ \rightarrow h^\pm l^+ l^\mp$ decays

Madalina Mironiuc

January 2020

Abstract

The Standard Model is currently thought as the best description of the Universe, however there are a few observations that contradict the predictions it makes, like the asymmetry of matter and antimatter in the Universe, dark matter and dark energy. Extended searches are made to discover new physics that can explain these observations, especially in the flavour sector which is considered to be the most promising in this area. This project is looking at the analysis of lepton flavour violating decays using LHCb data that was taken in 2017/2018. There already exists an analysis for the same decay channels for 2015/2016 data that set new limits for their branching ratios. The aim of this project is to improve on that analysis and use the tools introduced there to analyse 2017/2018 data and provide better limits on the branching ratios of the decay channels.

1 Introduction

The discovery of the Higgs boson in 2012 completed the Standard Model (SM) which is still the most accurate description of the universe that we have. However, there are a few observations that can not be explained using the mechanisms within in the SM: for example neutrino oscillations, the asymmetry of matter and antimatter in the universe and the nature of dark matter. There are different extensions to the SM proposed to explain these observations, the most known one being Supersymmetry (SUSY). The validity of these models is probed with new measurements and experiments at dedicated accelerators or investigating cosmic rays. Lately, searches for physics beyond the Standard Model have been focused on flavour physics and finding evidence of flavour violation in the lepton sector.

1.1 The Standard Model

The Standard Model is a gauge theory that has been successful in describing the fundamentals of the Universe. It provides reliable explanations and predictions for the unification of three of the fundamental forces: electromagnetic, strong and weak nuclear forces as physical manifestations of local gauge symmetries. Gravity is the remaining force to be included in what is known as a Grand Unified Theory. The symmetry of the SM is expressed in equation (1)

$$G_{SM} = SU(3)_C \times SU(2)_L \times U(1)_Y \quad (1)$$

Particle			Spin	Colour	Q_{EM}	Mass (GeV)		
u	c	t	$\frac{1}{2}$	(3)	$+\frac{2}{3}$	2.2×10^{-3}	1.275	173
d	s	b	$\frac{1}{2}$	(3)	$-\frac{1}{3}$	4.7×10^{-3}	0.095	4.18
e^-	μ^-	τ^-	$\frac{1}{2}$	(1)	-1	0.511×10^{-3}	0.105	1.78
ν_e	ν_μ	ν_τ	$\frac{1}{2}$	(1)	0	$< 2 \times 10^{-9*}$	(0 in SM)	
γ			1	(1)	0	0		
g			1	(8)	0	0		
$W^\pm$			1	(1)	± 1	80.4		
Z^0			1	(1)	0	91.2		
H			0	(1)	0	125		

Table 1: Properties of the particles in the Standard Model. The first two rows present the quark family, the next rows the lepton family and the last part presents the gauge bosons and the Higgs boson

where the group symmetries are each associated with a charge-like property that is intrinsic to particles. This symmetry is broken in the electro-weak sector through the Higgs spontaneous symmetry breaking mechanism becoming equation (2).

$$G_{SM} = SU(3)_C \times U(1)_{EM} \quad (2)$$

In the SM particles can be categorised as fermions, that interact with each other through the weak, strong and the electromagnetic force, and bosons that mediate these interactions. The fermions can be further divided into two families, each having three generations: the quark family and the lepton family containing the e^- , μ , τ and their corresponding neutrinos. For the gauge bosons, the photon is the mediator of the electromagnetic force, the charged $W^\pm$ and neutral Z^0 bosons mediate the weak interactions and the gluon mediates the strong interaction. The Higgs boson is responsible for giving fermions their mass; interactions with the Higgs field are called Yukawa interactions. A few properties of these particles can be found in table 1.1

1.2 Flavour physics

In the Standard Model fermions appear in three generations. Flavour physics describes the interactions that differentiate between these, the term flavour being used to describe copies of the same gauge representations, i.e. fields that have the same quantum numbers. [7]

The electromagnetic and strong interactions behave in the same way independently of the fermion flavours involved. Therefore they are known as flavour universal (or flavour blind) within the SM and in any of the proposed extensions to the model. The reason for this is because they correspond to exact gauge symmetries, conserving flavour. [6]

The weak interactions mediated by the Z^0 boson are considered universal in the Standard Model, but this is not true for other extensions of the model. The weak interactions mediated by the $W^\pm$ bosons are flavour violating. In fact these are the only processes where flavour violation is observed at the tree level and are known as flavour changing charged current(FCCC) processes, which mediate the nuclear beta decay. [8]

The Yukawa interactions are not flavour universal, but they are diagonal in the Standard model due to the fact that the Higgs mechanism is the only source for the mass of the particles and

the diagonalization of the mass matrix leads to the diagonalisation of the Yukawa interactions. However, as it was the case for the weak interactions, this is only true in the Standard Model. [8]

Flavour symmetry in the Standard Model is broken by the Yukawa couplings. The only term that allows for flavour violation within the model is the complex phase of the Cabbibo-Kobayashi-Maskawa matrix as explained in the next section. Flavour physics therefore describes the parts of the SM that depend on the Yukawa couplings, these being the only way to distinguish between the tree families of leptons and quarks.

1.3 CP violation and flavour violating mechanisms

CP is the symmetry that relates the particles with their anti-particles. If CP is not violated, the amplitude of the decay involving the particle should be the same as for the antiparticle, therefore CP violation can only appear in interference between two decay amplitudes. In the SM, CP violation comes only when there are three generations and all of them are involved. [6]

CP violation is closely related to Flavour Physics. In the SM the CKM complex phase is the one that allows for CP violation, but the measured rates are lower than what Baryogenesis suggests, therefore there must be other sources of CP violation. It is believed that flavour changing interactions might provide the answer for this discrepancy. [6]

Fermions can be involved in two types of interactions: gauge interactions, where two fermions coupling to a gauge boson and Yukawa interactions, having them couple to a scalar (in the SM this describes the coupling to the Higgs field). As a result of this they can be written in two bases: the interaction basis and the mass basis.

In the interaction basis, gauge interactions are universal and diagonal, their couplings are independent for each interaction which means that there are no gauge couplings between interactions eigenstates of the same generation, as it is the case for the strong and electromagnetic forces.

In the mass basis the Yukawa couplings are diagonal, but not universal. However the mass eigenstates have a well defined mass and this is actually measured in particle detectors. These interactions are related to the spontaneously broken symmetries and allow for fermion mixing. [7]

The SM flavour section arises from the Yukawa couplings as it can be seen in equation (3) :

$$\mathcal{L}_Y = (Y_u)_{ij} \bar{Q}_{Li} u_{Rj} \tilde{\Phi} + (Y_d)_{ij} \bar{Q}_{Li} d_{Rj} \Phi + (Y_e)_{ij} \bar{L}_{Li} e_{Rj} \Phi + h.c. \quad (3)$$

where i and j are running over the three generations so that the Y are 3×3 complex matrices. The fields are the following: Q_L and L_L are the left-handed quark doublets respectively lepton doublets, u_R , d_R and e_R are the right-handed quark singlets, respectively lepton singlets. Φ represents the Higgs field and $\tilde{\Phi}$ represents the conjugate Higgs field defined as equation (4):

$$\langle \phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \quad (4)$$

where v is the vacuum expectation value, $v \simeq 246$ GeV.

This equation does not contain a mechanism that gives mass to neutrinos, but the Yukawa matrices (and thus the fermion mass matrices) can be diagonalised by unitary rotations in the fields. Due to the unitarity of these transformations, they do not modify the kinetic forms and the neutral interactions, therefore the photon and Z^0 boson are flavour conserving. Couplings to the Higgs are proportional to mass matrix so these are also flavour conserving. But rotations induce flavour violation in the charged current interactions which are described by CKM matrix. [3]

The CKM matrix that correlates the interaction basis to the mass basis and allows for flavour changing decays in the standard model is described in equation (5)

$$V_{CKM} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}, \quad (5)$$

where $|V_{ij}|^2$ is the probability that a quark of flavour i will decay into a quark of flavour j . This leads to the matrix being unitary,

$$V_{CKM} V_{CKM}^\dagger = 1, \quad (6)$$

where the adjoint of this matrix $V_{CKM}^\dagger$ describes the interaction for the anti-particles.

Observations of CP violations in different decays have set the $V_{ij} \neq V_{ij}^*$ constraint which leads to the CKM matrix being reduced to three angles and a phase term which are measured experimentally. The main feature of the CKM matrix is the fact that the diagonal elements are approximately equal to one, but the off diagonal elements are much smaller corresponding to suppressed processes.

Another mechanism that contributes to flavour changing interactions is the flavour changing neutral current (FCNC) which is suppressed in the Standard Model at tree level and further, at loop level, by the GIM mechanism. Figure 1 presents the diagrams corresponding to FCNC mediating the flavour changing process $c \rightarrow ull$.

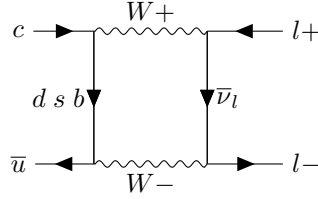


Figure1: Example of Flavour Changing Neutral Current (FCNC) process

A minimal extension of the SM is the introduction of a Higgs doublet [1], but the implications of this model is the fact that FCNC are generated at tree level, which has been experimentally excluded, therefore new theories are trying to propose methods of suppressing these processes. One of them is in the Minimal Supersymmetric Standard Model (MSSM) that introduces the $\mathbb{Z}_2$ symmetry and gives parity to the fermion fields that will cause them to only interact with one of the two Higgs doublet. Another solution is to impose the Minimal Flavour Violation (MFV) model requirements on the Yukawa couplings. [3]

2 Beyond the standard model measurements

All the models proposed in high energy physics have a number of free parameters that are determined using different measurements. If the predictions made using the first measured values correspond to new observations, then the model is correct. The SM has 19 free parameters out of which 13 are to flavour physics parameters (if we do not account for Dirac neutrino masses): the masses of the quarks and leptons (9) and the 3 angles and one complex phase of the CKM matrix. [6]

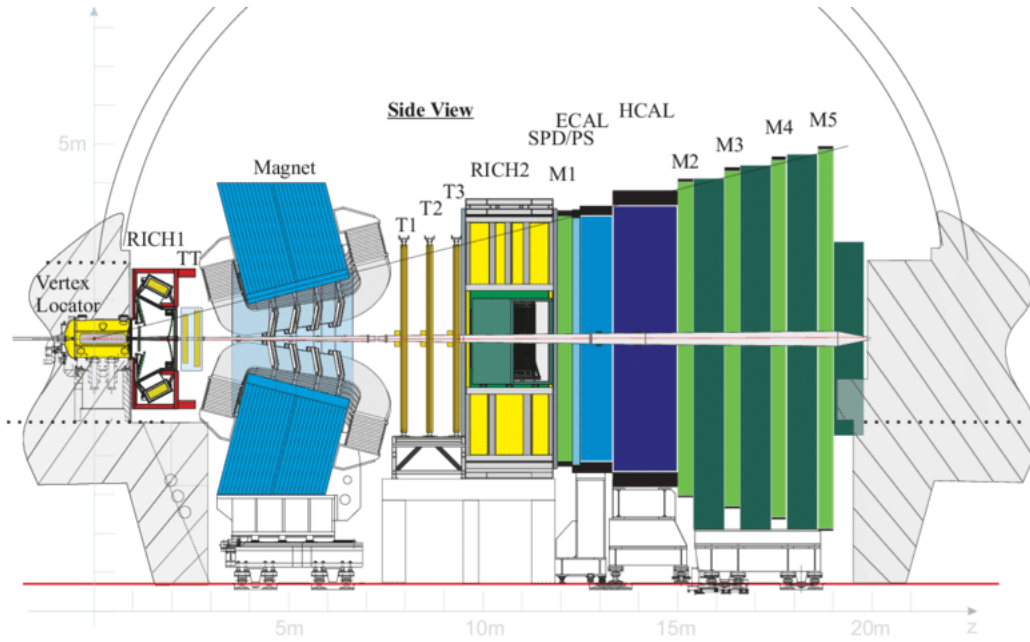


Figure 2: The layout of the LHCb detector [5]

There are two types of measurements: direct and indirect. The first refers to processes that take place at tree level and they are expected to hold almost model independent. Indirect searches are those that are described by loop diagrams. Their main advantages are the fact that loop processes allow the formation of particles that are more energetic than the initial colliding particles and that new physics will usually have a large effect on processes that are suppressed by the SM, mainly loop processes. [6]

3 The LHCb detector

The LHCb is one of the four main detectors at the Large Hadron Collider (LHC). It's a single-arm forward spectrometer covering pseudorapidity range $2 < \eta < 5$ especially designed to study interactions between particles containing b or c quarks. The main purpose of the experiment is measuring CP violation in decays allowed in the Standard Model, measuring decay rates for meson mixing and searching for physics beyond the Standard Model.

The components of the detector as they can be seen in figure are a vertex locator (VELO), silicon tracker, two ring imaging Cherenkov detectors (RICH), an electromagnetic calorimeter (ECAL), a hadronic calorimeter (HCAL) and five muon stations.

The vertex locator (VELO) is unique at the LHC as it can be placed very close to the proton beam (closest approach 7mm). This feature provides excellent primary vertex resolution and can accurately measure the flight distance of short-lived particles, for example the B mesons, which have a lifetime of $1.5 \times 10^{-12} s$.

The silicon tracker has two components: one placed after the magnet, the Tracker Turicensis and the other one placed downstream of the magnet, the Inner Tracker. They are both made of vertical silicon microstrip detectors and they measure the momentum of charged particles whose trajectories will be bent by the magnetic field.

The RICH detector contains a gas with a refractive index such that charged particles that pass through travel faster than the local speed of light and emit Cherenkov radiation. The velocity of the particle can be calculated using the angle of the emitted radiation. Combining this information to the one obtained from the tracker the mass of the particle can be identified.

The calorimeters measure the energy and position of the incident particles and provide a signal for the lowest LHCb trigger. The ECAL measures the energy of lighter particles, like electrons and photons, whereas the HCAL is designed to record the energy of the hadrons. This is especially useful in detecting neutral particles like neutrons and photons.

Muons can penetrate multiple surfaces and have a relatively long lifetime, which is why the muon detectors are the outermost part of particle detectors. This is also the reason why muons are extremely useful in analysis - they can be easily and unambiguously identified. The LHCb muon stations contain wire chambers filled with gases so that incoming muons interacting with the gas will cause the wire electrodes to detect a signal.

4 $D_{(s)}^+ \rightarrow h^\pm l^+ l^\mp$ analysis

This section will present the improvements made to the analysis methods used for $D_{(s)}^+ \rightarrow h^\pm l^+ l^\mp$ data, where h can be a kaon or a pion and l is a muon or an electron. The 2015/2016 data has already been analysed [2] and set new limits on the branching fractions for most of these decays. The aim of this project is to improve the existing analysis so that new limits can be set for all the channels for the 2016 data and set the tools and algorithms for analysing 2017/2018 data. The use of natural units ($c = \hbar = 1$) and inclusion of charge conjugate processes is assumed.

The analysis is performed on 28 decay channels. They have the same topology, but they occur through various processes. Eight of these decays are allowed in the Standard Model, through the Weak Annihilation process as seen in figure 2. Four of these are mediated by the $W^\pm$ and the other four through Flavour Changing Neutral Currents. The remaining 20 decays are Lepton Flavour Violating (LFV) and can occur via the an oscillating neutrino. The 12 of them that are also Lepton Number Violating (LNV) have been predicted by extensions of the Standard model that involve a Majorana neutrino as seen in figure 3. The branching ratios of these decays are too small to be measured, therefore any signal observed in these 20 channels would be conclusive evidence of beyond the Standard Model physics.

Table 4 lists all the decays that 28 decays channels analysed.

4.1 Data flow

In the detector one collision produces millions of events that have to pass through the detector and be saved as data. However that amount of information would be difficult to store and analyse, therefore the detector has hardware and software triggers, that determines what data point to keep. After the data was recorded it is reconstructed, i.e measurements from all the different parts of the detector are combined to fully described the processes that have taken place during that collision.

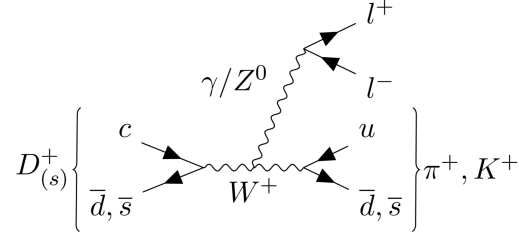


Figure 3: The Feynman diagram of the weak annihilation decays of the D_s^+ meson

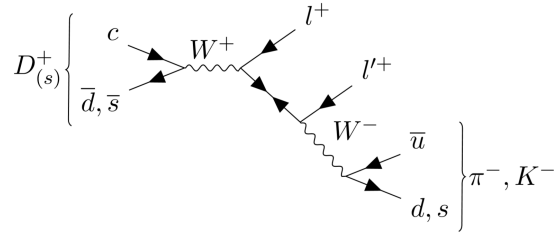


Figure 4: The Feynman diagram of the decays of the D_s^+ meson mediated by a Majorana neutrino that can result in LFV and LNV decays

Channel	Type
$D_{(s)}^+ \rightarrow \pi^+ e^+ e^-$	FCNC
$D_{(s)}^+ \rightarrow \pi^+ \mu^+ \mu^-$	FCNC
$D_{(s)}^+ \rightarrow K^+ e^+ e^-$	Weak annihilation DCS
$D_{(s)}^+ \rightarrow K^+ \mu^+ \mu^-$	Weak annihilation DCS
$D_{(s)}^+ \rightarrow \pi^+ e^+ \mu^-$	LFV
$D_{(s)}^+ \rightarrow \pi^+ e^- \mu^+$	LFV
$D_{(s)}^+ \rightarrow K^+ e^+ \mu^-$	LFV
$D_{(s)}^+ \rightarrow K^+ e^- \mu^+$	LFV
$D_{(s)}^+ \rightarrow \pi^- e^+ e^+$	LFV LNV
$D_{(s)}^+ \rightarrow \pi^- \mu^+ \mu^+$	LFV LNV
$D_{(s)}^+ \rightarrow K^- e^+ e^+$	LFV LNV
$D_{(s)}^+ \rightarrow K^- \mu^+ \mu^+$	LFV LNV
$D_{(s)}^+ \rightarrow \pi^- e^+ \mu^+$	LFV LNV
$D_{(s)}^+ \rightarrow K^- e^+ \mu^+$	LFV LNV

Table 2: The 28 decay channels that are used in this analysis and the type of processes through which they occur

The information is then stored and stripped so that the files used in analysis contain only the relevant data for the decay.

The data is then pre-processed where cuts are applied on different variables, for example the probability that the leptons are electrons or muons, the momentum of the final particles. The output obtained after applying these cuts is the input into the analysis described in the following.

The first step in the analysis is blinding. A big concern in particle physics is that the results are affected by statistical fluctuations which should be quantified by final uncertainties, but there are also unconscious biases that appear while conducting the analysis. The solution for this is to conduct a blind analysis which is achieved by hiding part of the data while the analysis is developed. Ideally the analysis will get rid of the existing background so that when it is applied to the whole data set the results have to be taken as they are, without further investigation.

For this analysis the selected channels are dominated by SM resonances $D_{(s)}^0 h^\pm R$, where the meson will decay into a kaon or a pion and a SM resonance $\rho, \omega \phi$ which will decay into a pair of electrons or muons. These decays can be used as normalisation channels, for this analysis the decays $D_{(s)}^+ \rightarrow \phi e^\pm \mu^\mp$ are the normalisation channels.

Given the complexity of this process and the fact that the project is focused on rare decays that have small branching fractions, the data is cross-checked using Monte Carlo (MC) samples. These are data points that have been obtained by simulating the detector response using specifications of the real detector as input. The main advantage is that MC can generate any number of decays which improve the analysis because it allows for exploring the data behaviour when these channels have a significant signal. However, to make sure that MC samples represent data properly they need to be reweighted to mirror the real data.

All the techniques presented before are meant to separate signal from background by applying rectangular cuts, i.e. for each variable there a maximum or minimum value is specified and only the data points above, respectively below these is considered signal. However, there are correlations between variables that can not be explored using this kind of cuts. The solutions suggested for this is to train a classifier and use its result as an additional cut to better separate the signal and background.

4.2 The classifier: Boosted decision trees

A classifier is an algorithm that predicts the class (signal or background) of a data point in a multidimensional space. The specific classifier that was chosen for this analysis is a boosted Binary Decision Tree (BDT). A binary decision tree applies successive rectangular cuts based on different criteria which means that more complex shapes are made in the phase space of the problem that will describe the data distribution more accurate. This is exemplified by figure 4.

These classifiers can be further improved by using a technique called boosting, where multiple BDT are combined and trained together to improve the classifier performance. They are trained in a supervised fashion, giving the algorithm labelled data and using the combination of learners that give the best separation and then apply the trained algorithm to the data that needs classification.

The classifier used is the XGBoost algorithm [4] with the same specifications as described in [2]. The channels for D^+ and D_s^+ are combined when training the classifier. The training signal data is constituted from the MC data for all the channels. The background training data is a sample of unphysical backgrounds where all the final states particles have the same charge.

The variables used to train the classifier have to be uncorrelated to the mass and descriptive enough as to offer a good separation between the two classes. The variables chosen are:

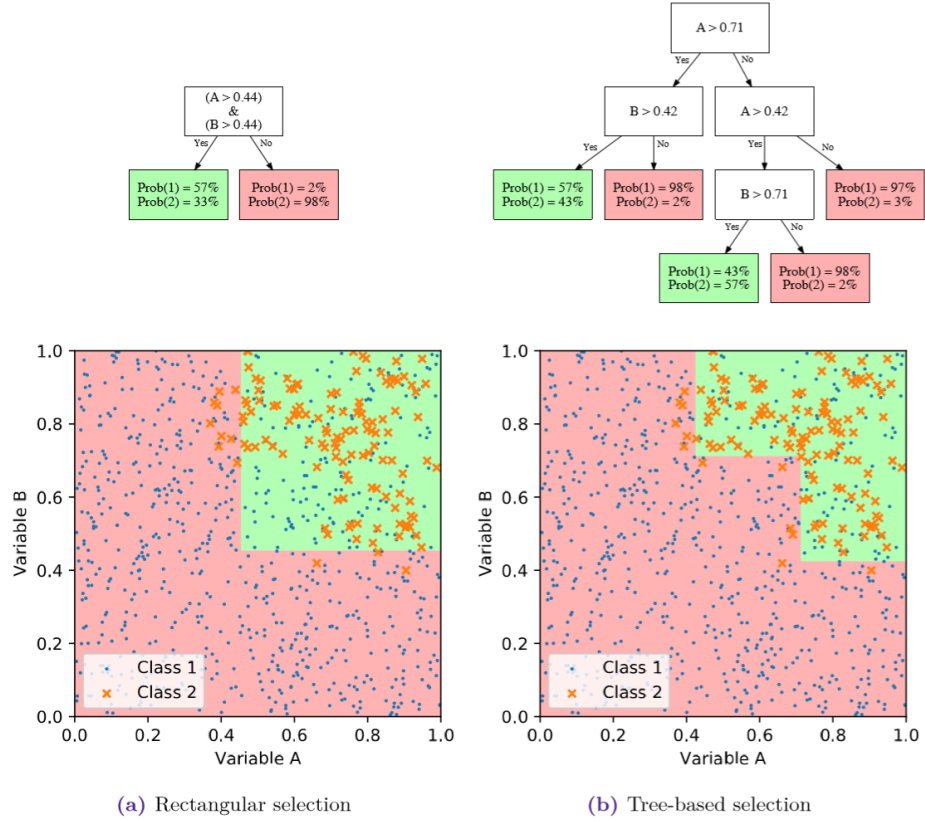


Figure 5: The difference between applying a rectangular cut to the data and using a Decision Tree to find the optimal separation between signal and background data [2]

Parameter	Value
max_depth	4
learning_rate	0.1
min_child_weight	10
colsample_bytree	0.5
gamma	0.1
scale_pos_weight	$\frac{N_{BKG}}{N_{Signal}}$
n_estimators	150

Table 3: The hyperparameters used when training the multivariate classifier [2]

- Primary vertex fit quality
- $D_{(s)}^+$ decay vertex fit quality per degree of freedom
- Pseudorapidity of the $D_{(s)}^+$ meson
- Flight distance of the $D_{(s)}^+$
- Impact parameter of² the final state particles
- Magnitude of each of the final state particle's three-momenta
- Impact parameter between the reconstructed $D_{(s)}^+$ and the primary vertex
- pT asymmetry in a 2 radian cone around the $D_{(s)}^+$ candidate
- Maximum distance of closest approach between all final state particles
- Angle between the reconstructed $D_{(s)}^+$ momentum and the line between the particles origin and decay vertices
- Reconstructed proper lifetime of the $D_{(s)}^+$

The hyperparameters that improve the classifier performance were chosen empirically and they can be found in table 4.2.

The classifiers performance is evaluated by comparing the obtained Receiver Operating Characteristic (ROC) curves. The ROC curve is a graphical representation of the discriminative power of the classifier. It plots the true positive rate versus the false positive rate for the test data. Classifiers that perform better will increase the area under this curve.

The selected classifier separates well the signal from the background data for most channels, but it was observed after the data was unblinded that there were some unexpected background data for three of the channels, which restricted the ability to establish better limits for their branching fractions. Therefore, the classifier had to be improved before it can be used to analyse 2017/2018 data.

Two things were suggested to improve the classifier: to add more training variables and to train the classifier using more data. Adding more variables could be problematic because it could lead to the classifier misidentifying statistical fluctuations as correlations between variables. On the other

hand, there are two types of background that can be added to further train the classifier: MC simulations of different backgrounds that can appear in the data and real background data that has been misidentified as signal.

For the purpose of this project there were two channels that have been used for both MC and real background data: $D_{(s)}^+ \rightarrow K\pi\pi$ and $D_{(s)}^+ \rightarrow \pi\pi\pi$ as they are considered as the predominant backgrounds found in the data, with $K\pi\pi$ having the greatest contribution.

The classifier was trained using different combinations of these files and the ROC curves for the different classifiers for each individual channel were plotted on the same graph. The combinations used were:

- Using the MC simulation for the $K\pi\pi$ channel, labelled as ‘ONE MC’
- Using the MC simulation for both the $K\pi\pi$ and $\pi\pi\pi$ channels, labelled as ‘MORE MC’
- Using the real data for the $K\pi\pi$ channel, labelled as ‘ONE BKG’
- Using the real data for both the $K\pi\pi$ and $\pi\pi\pi$ channels, labelled as ‘MORE BKG’
- Using the MC simulation and real data for the $K\pi\pi$ channel, labelled as ‘ONE MC, ONE BKG’
- Using the MC simulation for both the $K\pi\pi$ and $\pi\pi\pi$ channels and the real data for the $K\pi\pi$ channel, labelled as ‘MORE MC, ONE BKG’
- Using the MC simulation for the $K\pi\pi$ channel and the real data for both the $K\pi\pi$ and $\pi\pi\pi$ channels, labelled as ‘ONE MC, MORE BKG’
- Using the MC simulation and real data for both the $K\pi\pi$ and $\pi\pi\pi$ channels, labelled as ‘MORE MC, MORE BKG’

The ROC curves were plotted using the MC signal data, the real blinded data and the real background. The results for the channels $D_{(s)}^+ \rightarrow K^+e^-e^+$, $D_{(s)}^+ \rightarrow KK^+\mu^-e^+$, $D_{(s)}^+ \rightarrow K^+\mu^-\mu^+$ are showed in figures 5 - 7.

It can be seen that adding the real (misidentified) background will improve the classifier for the channels containing an electron but will not make a difference for the ones where both leptons are muons. This behaviour can be explained by the fact that muons are identified more reliable by the detector as they are the only particle that will penetrate all the other parts of the detector and end up in the muon chambers.

5 Future work

These were the only channels that the classifier was tested on. It seems that adding more real background data to the classifier is improving its performance, however future work will involve testing the classifier on all the channels and also trying different variables that can increase its ability to differentiate between signal and background data. The best configuration for the classifier will then be included in the analysis to be used for the 2017/2018 data and offer better results for the channels where the previous analysis did not account for all the background.

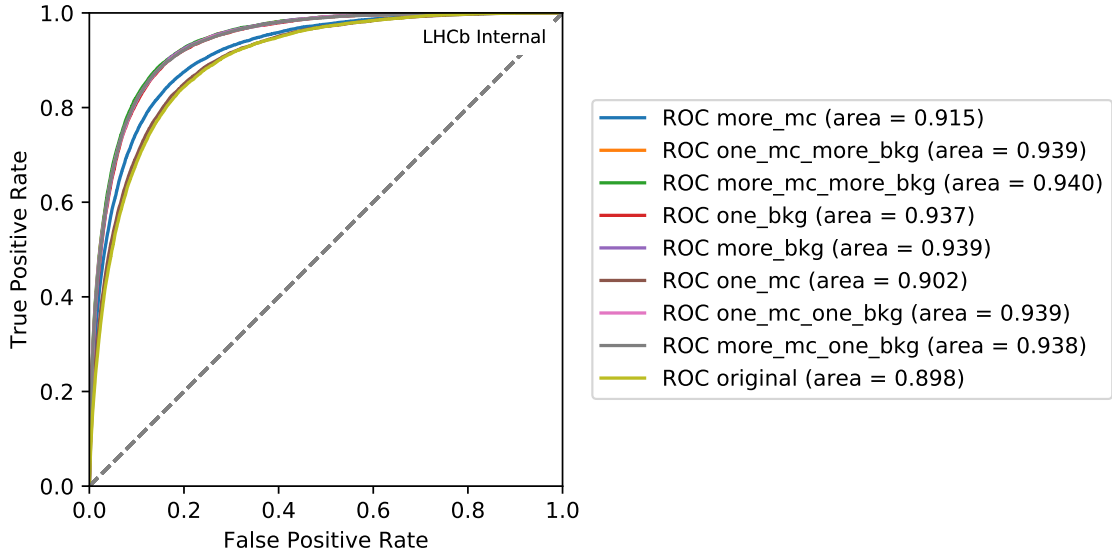


Figure 6: The ROC curves comparing the different classifiers for the $D_{(s)}^+ \rightarrow K^+ e^- e^+$ decay channel

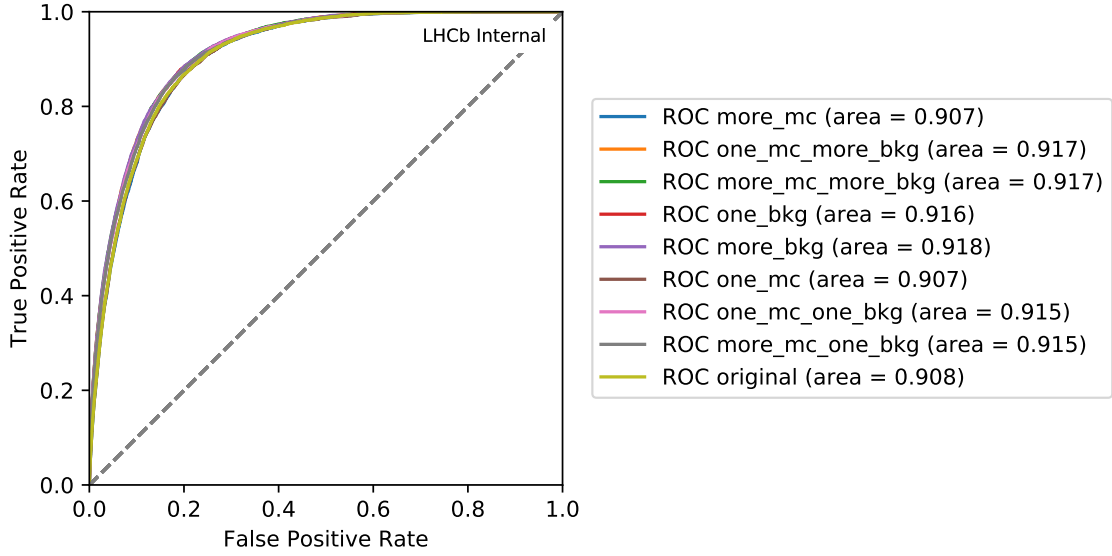


Figure 7: The ROC curves comparing the different classifiers for the $D_{(s)}^+ \rightarrow K^+ \mu^- e^+$ decay channel

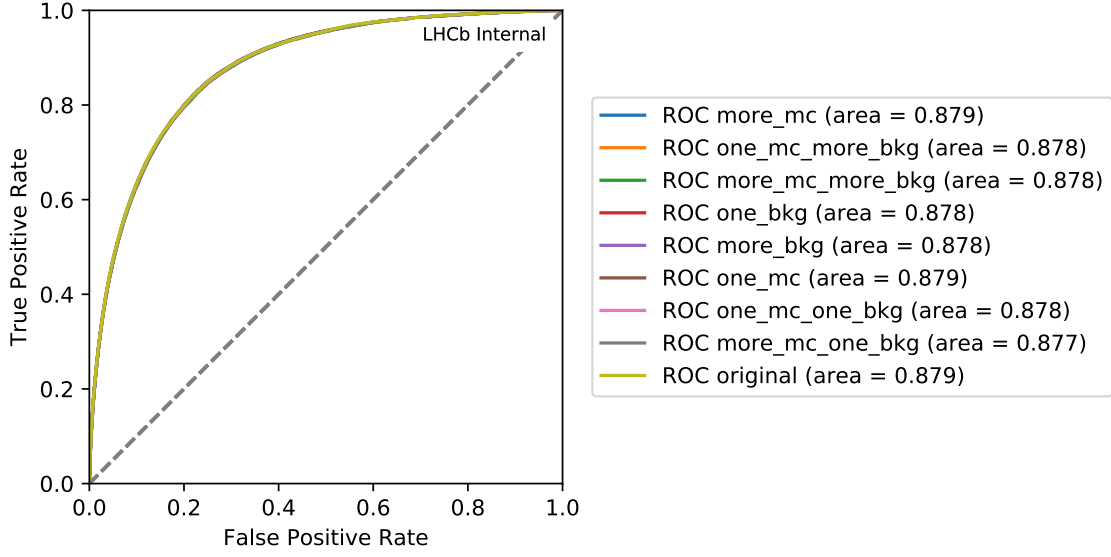


Figure 8: The ROC curves comparing the different classifiers for the $D_{(s)}^+ \rightarrow K + \mu^- \mu^+$ decay channel

References

- [1] Gustavo Castelo Branco, PM Ferreira, L Lavoura, MN Rebelo, Marc Sher, and Joao P Silva. Theory and phenomenology of two-higgs-doublet models. *Physics reports*, 516(1-2):1–102, 2012.
- [2] Chris Burr. Doctoral thesis: Searching for rare charm decays, performing alignment studies and improving the analysis ecosystem in hep. 2019.
- [3] Lorenzo Calibbi and Giovanni Signorelli. Charged lepton flavour violation: an experimental and theoretical introduction. *arXiv preprint arXiv:1709.00294*, 2017.
- [4] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pages 785–794, 2016.
- [5] LHCb Collaboration. Lhcb detector performance. *International Journal of Modern Physics A*, 30(07):1530022, 2015.
- [6] Yuval Grossman. Introduction to flavour physics. In *LHC Phenomenology*, pages 35–80. Springer, 2015.
- [7] Yosef Nir. Probing new physics with flavor physics (and probing flavor physics with new physics). *arXiv preprint arXiv:0708.1872*, 2007.
- [8] Yosef Nir. Flavour physics and cp violation. *arXiv preprint arXiv:1010.2666*, 2010.